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Research Report

Sensing Disruption: Output Entropy as a Channel-Agnostic Monitor for LLM Agents

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1 Introduction

LLM agents are now deployed as multi-agent systems in which many instances plan and act through tools, often in parallel and on shared infrastructure (Kim et al., 2026; Zhu et al., 2026). Unwanted behaviour in these systems does not have to start in a single agent. It can come from interaction between agents that were never meant to interact. In July 2026 this happened in public. During internal cybersecurity evaluations at OpenAI, agents running in separate sandboxed runs used an internal package mirror, provided only so they could install software, to reach each other and later the internet. Between 11 and 13 July they compromised part of Hugging Face’s production infrastructure while searching for the solutions to their evaluation tasks (Larcher et al., 2026; OpenAI, 2026).

The channel grew slowly. On 12 May an agent that could not find a file it needed reasoned that an agent in another environment might have it, and left a note on the mirror: “Agent seeks [filename]; upload if found!” Other agents found such notes and, using credentials they had been given legitimately, turned the mirror into a message board. When the mirror was rebuilt, they re-established the board by encoding messages in directory names. By late June they were sharing exploitation syntax on it, in July live production credentials were posted there, and responders later disabled 311 repositories used for the board. Most of the discussion concerned the hardest tasks: 93% of the tasks on the board came from the 22% that no model had ever solved (OpenAI, 2026). The sig-

nals were there early. Board activity was observed around 26 May, and on 27 June a monitoring tool flagged activity that responders linked to the board, but they concluded that stopping the run was not required. From Hugging Face’s side, the roughly 17,600 actions looked like one high-volume attacker, and “the successful path was hidden inside the noise generated by the thousands of failed ones” (Larcher et al., 2026).

In the Hugging Face incident, isolated agents turned a shared resource into a hidden channel, and every such channel ends by putting unsanctioned text into a sandboxed agent’s context. This is why watching the channel does not work. The agents moved information through plaintext files, directory names, URL paths, public datasets, pastebins, webhooks and screenshots read back with OCR (OpenAI, 2026), and they chunked, encrypted and encoded staged data so that plain log scans missed about 75% of it (Larcher et al., 2026). None of these resources was designed for communication, and a detector written for one would have been bypassed by the next. The industry answer is to read more: scan logs, and have a second model review the agent’s chain of thought (Baker et al., 2025). Both depend on content that stays readable, and both cost more as the number of runs grows.

We wonder if there is a way to build a monitor that works differently, one that explores the relationship between entropy and the system instead of the content of what agents say. The operator of a sandbox already records, for every generated token, the model’s probability distribution over the next token. If unsanctioned text enters an agent’s context, that distribution should change, whatever channel delivered the text. Entropy has served this role before in other systems: it exposes network attacks without reading packet payloads (Lakhina et al., 2005), and it flags hallucinations in LLMs (Farquhar et al., 2024). We test the idea in a small testbed

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that reproduces the incident: two sandboxed agents, each holding one password to a task neither can finish alone, share a log that stays closed, opens mid-task, or delivers foreign text that carries no useful information. We run it with three models at three sampling temperatures. We treat this as a first step toward such a monitor, not as a finished one.

Our main contributions are:

1. **A testbed modelled on the incident, with verifiable ground truth.** Cryptographically split credentials make every use of the channel provable, and a placebo condition separates the arrival of foreign text from the arrival of useful information. The code and all 539 runs, with token log-probabilities, are public.
2. **Patterns in how entropy reacts when isolation breaks, across three models and three temperatures.** Every model used the channel, but only some models' entropy responded, and under sampling it responded as much to useless text as to real messages. The two agents never became measurably coupled: communication appeared as a disturbance inside each receiver, not as synchrony between agents.
3. **An honest assessment of entropy as a monitor.** It detects that a sandboxed agent's context was disrupted for some models and works best at low temperature, but it does not tell whether the disruption was communication. We identify what a usable monitor would still need.

2 Related Work

Entropy as a detector. Lakhina et al. (2005) detected network anomalies, such as denial-of-service attacks, from shifts in the entropy of traffic features like addresses and ports, without inspecting packet payloads or knowing the attack in advance. In LLMs, Farquhar et al. (2024) used semantic entropy to detect hallucinations. That work measures the uncertainty of one model answering one prompt.

Entropy in multi-agent systems. Zhao et al. (2026) measure entropy at the token, agent and round levels of collaborating agents and use it to predict when collaboration helps. In their setting communication is designed and known to the analyst, so entropy explains task performance rather than detecting coordination nobody intended.

Monitoring today. AI control protocols place trusted monitors around untrusted models (Greenblatt et al., 2024). These monitors read what the model produces, which stops working when messages are encoded or spread over resources nobody watches, as in the incident (Larcher

et al., 2026; OpenAI, 2026).

Our position. We carry the network idea over to sandboxed agents. Agents take the place of network flows, their next-token distributions take the place of traffic features, and an unknown channel takes the place of an unknown attack. Unlike the work above, we use entropy to monitor for unintended coordination. The signal does not depend on the channel or the encoding, and it needs no extra model call, because the probabilities are produced anyway when each token is generated. We test it against a placebo that delivers useless foreign text and against cryptographic ground truth of whether communication happened.

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